Supplementary Material: Recording-Grouped Bearing Fault Diagnosis under Directional Operating-Condition Shifts on the MCC5 Electric-Drive Benchmark

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1 Preprocessing and Split Audits

All final neural and fusion runs use frozen source-recording roles. No source recording or window crosses training, validation, and test roles, and no window-level validation fallback is used. Standardization statistics are fitted on training data only. The audit contains 1425 checks and all pass.

Table S1: Normalization audit by publication-facing input definition and code identifier.

Publication label	Code identifier	Audited	Pass	Fail
Strict-254 scalar-branch fusion	`full_engineered_features_254`	1040	1040	0
Legacy-262 complete branch	`full_multisource_fusion`	268	268	0
Raw CNN/TCN/Transformer, 8,192 samples	`raw_8192`	24	24	0
Raw CNN length sensitivity, 12,800 samples	`raw_length_12800`	6	6	0
Raw CNN length sensitivity, 8,192 samples	`raw_length_8192`	6	6	0
Vibration-current; no scalars	`vibration_current`	7	7	0
Vibration-current + auxiliary-26	`vibration_current_auxiliary_only`	32	32	0
Vibration-current + auxiliary-context-28	`vibration_current_rpm_load`	34	34	0
Vibration-current + RPM/load	`vibration_current_rpm_load_only`	8	8	0

2 Raw-File Provenance and Acquisition-Session Audit

The official raw MCC5 copy was audited without redistributing signal samples. Each of the 84 catalogued recordings resolved to one CSV file with 1,152,000 data rows and nine columns (time plus eight measured channels). Every file has a distinct SHA-256 digest. One timestamp is shared by three filenames; streaming pairwise comparison shows a common time vector but different signal values for all three pairs. These checks exclude byte-identical and pointwise-identical signal files within the repeated-timestamp group, but they do not exclude transformed similarity or establish independent physical bearings or acquisition sessions.

Table S2: Compact raw-file identity audit.

Audit item	Result
Catalogued/resolved recordings	84 / 84
Distinct SHA-256 digests	84
Exact duplicate digest groups	0
Repeated timestamp groups	1
Files in repeated group	3
Pairwise signal-identical comparisons	0 of 3
Raw samples redistributed	No

The recordings occur on five acquisition dates, and class composition is uneven across those dates (Table S3). An in-sample date-majority mapping agrees with 61/84 labels (0.7262); Cramer's V is 0.7714 and normalized mutual information is 0.5989. These values are descriptive association measures computed on the same 84 recordings. They are not cross-validated diagnostic performance and do not show that the trained models use timestamp metadata. They indicate that recording grouping cannot exclude session-related nuisance structure in the measured signals.

Table S3: Acquisition-date and publication-class counts. Dates are parsed from public recording identifiers.

Date	Healthy	Inner race	Outer race	Rolling element
250702	12	0	0	0
250704	0	11	0	0
250707	0	12	11	24
250708	0	0	12	0
250821	0	0	2	0

3 Partition and Session-Proxy Controls

The direct partition control crosses random versus source-recording-grouped assignment with overlapping versus nonoverlapping windows. The XGBoost model seed is fixed at 42, and split seeds 42–51 define the repetitions. Random assignment places windows from all 84 source recordings in both training and test roles, even after overlapping windows are removed. Recording-level metrics are therefore undefined for the random protocols because no recording is held out intact.

Table S4: Direct partition comparison across ten parallel split seeds. Variability is split variability, not optimization-seed variability.

Protocol	Train/test source overlap	Window macro-F1	Window range	Recording macro-F1
Random, overlapping windows	84	0.9976 ± 0.0009	0.9963–0.9994	Not valid
Random, nonoverlapping windows	84	0.9953 ± 0.0022	0.9907–0.9981	Not valid
Recording-grouped, overlapping windows	0	0.9777 ± 0.0356	0.9092–0.9992	0.9829 ± 0.0361
Recording-grouped, nonoverlapping windows	0	0.9765 ± 0.0358	0.9075–0.9992	0.9829 ± 0.0361

On 250707, 47 recordings cover inner-race, outer-race, and rolling-element faults but no healthy class. Severity is not balanced across these classes: inner-race recordings are low severity, outer-race recordings are high severity, and rolling-element recordings include both. Random forest and XGBoost both achieve 1.0000 recording macro-F1 under mean-probability aggregation and majority vote across all ten recording-grouped splits. This establishes within-date separability for those three fault classes only; it is not a four-class or severity-controlled validation.

Two binary controls additionally fix both date and severity. The high-severity task compares 12 rolling-element with 11 outer-race recordings; the low-severity task compares 12 rolling-element with 12 inner-race recordings. Both models and both recording-level aggregation rules achieve 1.0000 macro-F1 and worst-class recall across split seeds 42–51 (Table S5). Each repetition tests only three high-severity or four low-severity recordings, however, and repeatedly partitions the same finite pool. These controls support within-date, severity-matched fault-location separability for the observed contrasts without establishing specimen or session independence.

Table S5: Same-date, severity-matched binary controls over ten recording-grouped split seeds. Values are recording-level mean ± standard deviation; mean-probability aggregation is shown because majority vote is identical.

Severity-matched task	Model	Test records/split	Accuracy	Macro-F1	Worst recall
H: rolling-element vs. outer-race	Random forest	3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
H: rolling-element vs. outer-race	XGBoost	3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
L: rolling-element vs. inner-race	Random forest	4	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
L: rolling-element vs. inner-race	XGBoost	4	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

Metadata-only baselines were fitted on each protocol's training recordings and evaluated on its test recordings. The operating-context-only baseline is weak, while acquisition date alone is predictive because date and class composition are associated. Combining date and operating context changes only the high-speed protocol materially (Table S6). Date is not an input to the diagnostic models; this control quantifies a possible shortcut in the dataset structure rather than model reliance on timestamp metadata.

Table S6: Training-only logistic-regression metadata controls at recording level.

Metadata	Protocol	Macro-F1	Worst recall	Accuracy
Date only	Source-recording	0.7560	0.3333	0.7500
Date only	Cross-mode	0.7723	0.5000	0.7381
Date only	Cross-load	0.7500	0.5000	0.7143
Date only	Cross-RPM	0.7028	0.4286	0.6786
Operating context only	Source-recording	0.0000	0.0000	0.0000
Operating context only	Cross-mode	0.1818	0.0000	0.2857
Operating context only	Cross-load	0.1818	0.0000	0.2857
Operating context only	Cross-RPM	0.1111	0.0000	0.2857
Date + operating context	Source-recording	0.7560	0.3333	0.7500
Date + operating context	Cross-mode	0.7723	0.5000	0.7381
Date + operating context	Cross-load	0.7500	0.5000	0.7143
Date + operating context	Cross-RPM	0.7739	0.4286	0.7500

Acquisition-date predictability was also tested within the 25 outer-race recordings, holding fault location fixed. However, severity and date are strongly coupled: all 12 low-severity recordings occur on 250708, while high-severity recordings occur on 250707 (11) or 250821 (2). Ten repeated source-grouped folds were used. A training-fold severity-majority rule reaches 0.9199 mean accuracy and 0.6387 macro-F1, exceeding the best signal-model accuracy and macro-F1 in Table S7. Neither approach recovers the rare date. Consequently, this experiment is interpreted as date–severity composite predictability rather than independent evidence of a session signature.

Table S7: Outer-race date–severity composite prediction across ten repeated source-grouped folds. Values are mean ± standard deviation.

Feature set	Model	Accuracy	Macro-F1	Worst-date recall
All 254 features	Random forest	0.7692 ± 0.1594	0.5223 ± 0.1230	0.0000
All 254 features	XGBoost	0.8154 ± 0.1088	0.5627 ± 0.0809	0.0000
All signals without order	Random forest	0.8468 ± 0.1140	0.5857 ± 0.0837	0.0000
All signals without order	XGBoost	0.8635 ± 0.0775	0.5995 ± 0.0547	0.0000
Vibration/current without order	Random forest	0.8231 ± 0.1375	0.5975 ± 0.1690	0.1000
Vibration/current without order	XGBoost	0.8391 ± 0.1098	0.5799 ± 0.0816	0.0000
Severity only	Training-fold majority rule	0.9199 ± 0.0034	0.6387 ± 0.0025	0.0000

4 Classical Source-Recording Aggregation

For each primary classical protocol, window probabilities were averaged within source recording before classification; majority vote was evaluated separately. Class-stratified bootstrap resampled complete recordings 2,000 times. These are conditional empirical resampling intervals for the finite recordings in the listed test set, not population intervals for future bearings or acquisition sessions. The high-speed limitation remains visible after aggregation, including zero worst-class recall under majority vote.

Table S8: Classical source-recording results under the four predefined protocols. Intervals are conditional 95% empirical intervals obtained by class-stratified resampling of the listed test recordings for mean-probability macro-F1.

Protocol	Model	Recordings	Mean-probability macro-F1	95% interval	Majority-vote macro-F1	Mean-probability worst recall
Predefined source-recording	XGBoost	12	1.0000	[1.0000, 1.0000]	1.0000	1.0000
Cross-mode	Random forest	42	0.9791	[0.9365, 1.0000]	0.9791	0.9167
Cross-load	Random forest	42	1.0000	[1.0000, 1.0000]	1.0000	1.0000
1000/2000 to 3000 rpm	Random forest	28	0.7020	[0.6355, 0.8000]	0.6444	0.1111

The degenerate $[1.0000, 1.0000]$ intervals occur because every class-stratified resample of the current finite test recordings is classified perfectly by the listed fitted model. They do not imply zero uncertainty for future physical bearings, installations, or acquisition sessions.

Table S9: Complete source-recording mean-probability matrix for five classical models and four predefined protocols. The model random seed is fixed at 42; this table compares model definitions rather than optimization-seed variability.

Protocol	Model	Recordings	Accuracy	Macro-F1	Worst-class recall
Predefined source-recording	Logistic regression	12	1.0000	1.0000	1.0000
Predefined source-recording	RBF-SVM	12	1.0000	1.0000	1.0000
Predefined source-recording	Random forest	12	1.0000	1.0000	1.0000
Predefined source-recording	MLP	12	1.0000	1.0000	1.0000
Predefined source-recording	XGBoost	12	1.0000	1.0000	1.0000
Cross-mode	Logistic regression	42	0.9524	0.9583	0.9167
Cross-mode	RBF-SVM	42	0.9762	0.9791	0.9167
Cross-mode	Random forest	42	0.9762	0.9791	0.9167
Cross-mode	MLP	42	0.9524	0.9583	0.9167
Cross-mode	XGBoost	42	0.9762	0.9791	0.9167
Cross-load	Logistic regression	42	1.0000	1.0000	1.0000
Cross-load	RBF-SVM	42	0.6905	0.5458	0.0000
Cross-load	Random forest	42	1.0000	1.0000	1.0000
Cross-load	MLP	42	1.0000	1.0000	1.0000
Cross-load	XGBoost	42	0.9762	0.9791	0.9167
1000/2000 to 3000 rpm	Logistic regression	28	0.5000	0.3583	0.0000
1000/2000 to 3000 rpm	RBF-SVM	28	0.2857	0.1585	0.0000
1000/2000 to 3000 rpm	Random forest	28	0.7143	0.7020	0.1111
1000/2000 to 3000 rpm	MLP	28	0.5000	0.3759	0.0000
1000/2000 to 3000 rpm	XGBoost	28	0.8214	0.6667	0.0000

The complete matrix shows that the high-speed boundary is not specific to the preselected random-forest row: all five classical models lose at least one class completely or nearly completely at source-recording level. Conversely, the cross-load RBF-SVM failure shows why reporting only one selected model per protocol is insufficient.

Table S10: Source-recording mean-probability results over ten repeated grouped source-recording partitions. The model seed is fixed at 42; variability is caused by partition membership.

Model	Macro-F1 mean ± SD	Macro-F1 range	Worst recall mean ± SD
Random forest	0.9743 ± 0.0414	0.9143–1.0000	0.9000 ± 0.1610
XGBoost	0.9829 ± 0.0361	0.9143–1.0000	0.9333 ± 0.1405

5 Complete Fusion-Input Evidence

The stored 26-field auxiliary definition contains 24 varying torque/key-phase waveform summaries and two modality-presence indicators that are constant in this complete-modality subset; the 28-field configuration additionally includes RPM and load. These two definitions were isolated after the 3000-rpm holdout had been inspected and are therefore post-hoc exploratory throughout this supplement. Their numerical detail is disclosed for transparency, not as confirmatory model selection evidence. The strict 254-feature and legacy 262-field branches are retained to document input-definition sensitivity.

Table S11: Complete fusion-input results across the four recording-grouped protocols. Values for source-recording, cross-mode, and cross-load are window-level macro-F1 means and standard deviations across three optimization seeds. The focused 1000/2000-to-3000-rpm column uses five optimization seeds. Auxiliary-26 and auxiliary-context-28 were identified after inspection of the same 3000-rpm holdout and are therefore post-hoc exploratory rather than confirmatory comparisons.

Configuration	Scalars	Source-recording	Cross-mode	Cross-load	1000/2000 to 3000 rpm
Predefined matched configurations					
Vibration-current; no scalars	0	0.8991 ± 0.0029	0.9417 ± 0.0062	0.8286 ± 0.0477	0.7081 ± 0.0986
Vibration-current + RPM/load	2	0.8455 ± 0.0375	0.9542 ± 0.0144	0.8309 ± 0.0885	0.7124 ± 0.0529
Strict-254 scalar-branch fusion	254	0.9939 ± 0.0011	0.9723 ± 0.0046	0.9299 ± 0.0350	0.4001 ± 0.0813
Legacy-262 complete branch	262	0.9943 ± 0.0012	0.9660 ± 0.0023	0.9571 ± 0.0266	0.4557 ± 0.0784
Post-hoc exploratory configurations on the same holdout					
Vibration-current + auxiliary-26*	26	0.9940 ± 0.0013	0.9820 ± 0.0086	0.9109 ± 0.0294	0.9162 ± 0.0245
Vibration-current + auxiliary-context-28*	28	0.9919 ± 0.0016	0.9884 ± 0.0154	0.9275 ± 0.0349	0.9095 ± 0.0284

* Post-hoc configurations identified after inspection of the same 3000-rpm holdout.

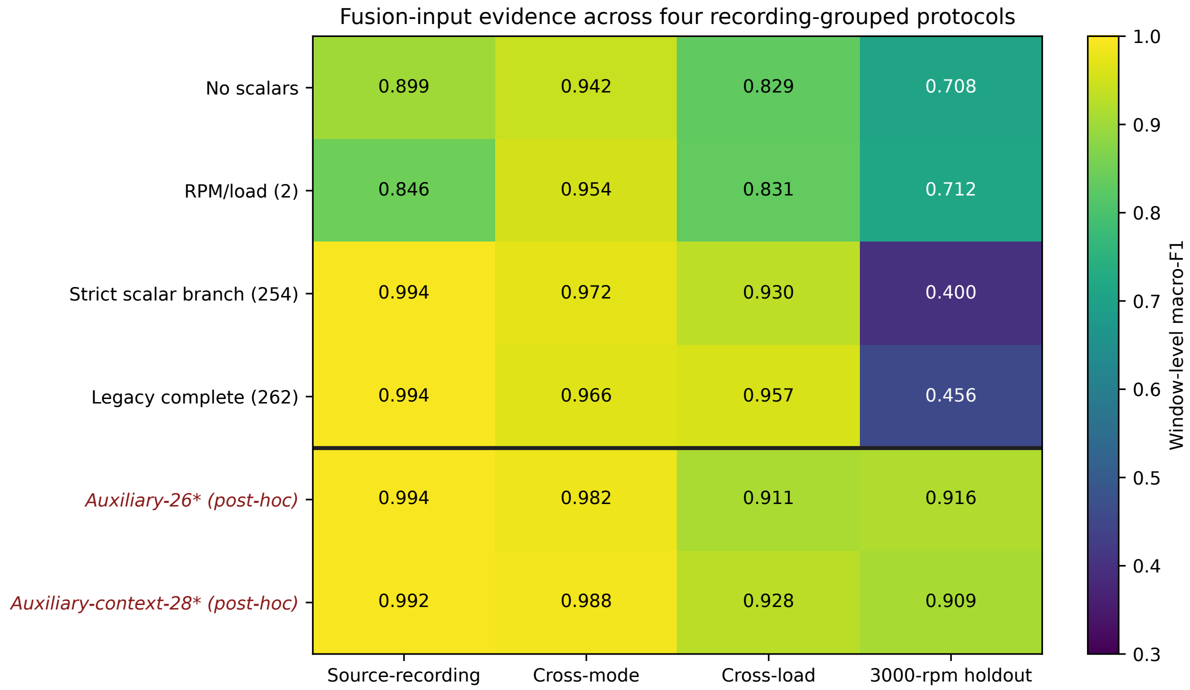

Figure S1: Complete four-protocol fusion-input comparison. The horizontal separator distinguishes predefined matched configurations from auxiliary-26* and auxiliary-context-28*, which are post-hoc exploratory configurations selected after inspection of the same 3000-rpm holdout. The displayed color range is 0.30–1.00.

5.1 Focused 3000-rpm evidence

Table S12: Per-seed fusion-input results for the 3000-rpm holdout. The auxiliary-26 and auxiliary-context-28 rows are post-hoc exploratory on this same holdout.

Configuration	Scalars	Seed	Macro-F1	Worst recall
Predefined matched configurations				
Vibration-current; no scalars	0	42	0.5599	0.2451
Vibration-current; no scalars	0	43	0.8027	0.6536
Vibration-current; no scalars	0	44	0.7366	0.4085
Vibration-current; no scalars	0	45	0.7792	0.3715
Vibration-current; no scalars	0	46	0.6623	0.4832
Vibration-current + RPM/load	2	42	0.7368	0.4714
Vibration-current + RPM/load	2	43	0.6231	0.2793
Vibration-current + RPM/load	2	44	0.7605	0.4874
Vibration-current + RPM/load	2	45	0.7116	0.3827
Vibration-current + RPM/load	2	46	0.7299	0.5154
Strict-254 scalar-branch fusion	254	42	0.5236	0.0223
Strict-254 scalar-branch fusion	254	43	0.3906	0.0279
Strict-254 scalar-branch fusion	254	44	0.4279	0.0321
Strict-254 scalar-branch fusion	254	45	0.3166	0.0251
Strict-254 scalar-branch fusion	254	46	0.3419	0.0209
Legacy-262 complete branch	262	42	0.4965	0.0251
Legacy-262 complete branch	262	43	0.5311	0.1648
Legacy-262 complete branch	262	44	0.4592	0.1243
Legacy-262 complete branch	262	45	0.3249	0.0223
Legacy-262 complete branch	262	46	0.4666	0.0880
Post-hoc exploratory configurations on the same holdout				
Vibration-current + auxiliary-26*	26	42	0.9121	0.8059
Vibration-current + auxiliary-26*	26	43	0.9431	0.8765
Vibration-current + auxiliary-26*	26	44	0.9274	0.8715
Vibration-current + auxiliary-26*	26	45	0.9209	0.8038
Vibration-current + auxiliary-26*	26	46	0.8773	0.7318
Vibration-current + auxiliary-context-28*	28	42	0.8819	0.6969
Vibration-current + auxiliary-context-28*	28	43	0.8820	0.6997
Vibration-current + auxiliary-context-28*	28	44	0.9429	0.8839
Vibration-current + auxiliary-context-28*	28	45	0.9336	0.8080
Vibration-current + auxiliary-context-28*	28	46	0.9070	0.8233

* Post-hoc configurations identified after inspection of the same 3000-rpm holdout.

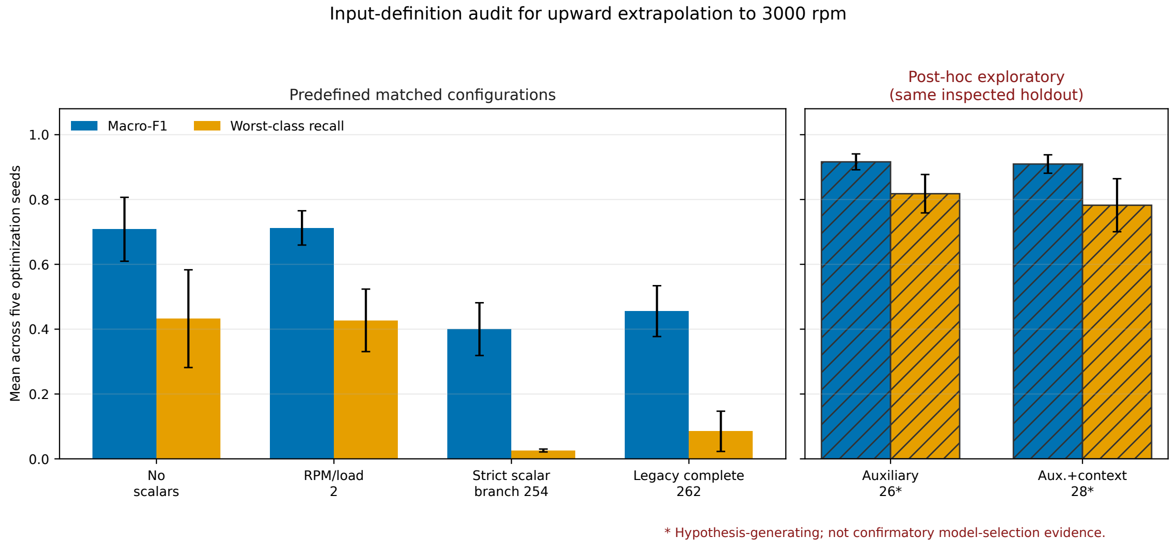

Figure S2: Focused fusion-input audit for upward extrapolation to 3000 rpm. The left panel contains predefined matched configurations; the separate hatched right panel contains post-hoc exploratory configurations identified on the same holdout. Error bars denote standard deviations across five optimization seeds. The right-panel comparison is hypothesis-generating rather than confirmatory.

Table S13: Recording-level mean-probability aggregation for all fusion-input definitions on the 3000-rpm holdout. Means and standard deviations are calculated across optimization seeds after probabilities have first been aggregated within each recording.

Configuration	Recordings	Recording macro-F1 mean ± SD	Recording worst recall mean ± SD	Evidence status
Vibration-current; no scalars	28	0.7122 ± 0.1238	0.4000 ± 0.2054	Matched protocol
Vibration-current + RPM/load	28	0.7306 ± 0.0917	0.4000 ± 0.1369	Matched protocol
Strict-254 scalar-branch fusion	28	0.3660 ± 0.1254	0.0000 ± 0.0000	Matched protocol
Legacy-262 complete branch	28	0.3829 ± 0.0917	0.0000 ± 0.0000	Matched protocol
Vibration-current + auxiliary-26*	28	0.9435 ± 0.0264	0.8361 ± 0.0669	Post-hoc exploratory
Vibration-current + auxiliary-context-28*	28	0.9368 ± 0.0513	0.8278 ± 0.1439	Post-hoc exploratory

* Post-hoc exploratory configuration identified after inspection of the same 3000-rpm holdout.

6 Severity and Per-Recording Probability Audits

The severity analysis is a descriptive, low-cost audit derived from saved source-recording predictions. It was not used for model selection. High- and light-severity fault recordings are evaluated separately; healthy recordings remain ungraded.

Table S14: Severity-stratified recording-level results on the 3000-rpm holdout. Severity is inferred from the public recording identifier; healthy recordings have no severity grade. Values are means and standard deviations across optimization seeds.

Configuration	Severity	Recordings	Accuracy mean ± SD	Fault macro-F1 mean ± SD	Evidence status
Vibration-current; no scalars	High	12	0.6333 ± 0.1728	0.5771 ± 0.2166	Matched protocol
Vibration-current; no scalars	Light	12	0.8667 ± 0.1264	0.8631 ± 0.1303	Matched protocol
Vibration-current; no scalars	Healthy (ungraded)	4	0.6000 ± 0.3354	Not applicable (single class)	Matched protocol
Vibration-current + RPM/load	High	12	0.6667 ± 0.1768	0.6272 ± 0.2172	Matched protocol
Vibration-current + RPM/load	Light	12	0.9333 ± 0.0373	0.9323 ± 0.0379	Matched protocol
Vibration-current + RPM/load	Healthy (ungraded)	4	0.5000 ± 0.2500	Not applicable (single class)	Matched protocol
Strict-254 scalar-branch fusion	High	12	0.4000 ± 0.1369	0.3600 ± 0.1710	Matched protocol
Strict-254 scalar-branch fusion	Light	12	0.7500 ± 0.1559	0.6929 ± 0.1965	Matched protocol
Strict-254 scalar-branch fusion	Healthy (ungraded)	4	0.0000 ± 0.0000	Not applicable (single class)	Matched protocol
Vibration-current + auxiliary-26*	High	12	0.8500 ± 0.0697	0.8474 ± 0.0725	Post-hoc exploratory
Vibration-current + auxiliary-26*	Light	12	1.0000 ± 0.0000	1.0000 ± 0.0000	Post-hoc exploratory
Vibration-current + auxiliary-26*	Healthy (ungraded)	4	1.0000 ± 0.0000	Not applicable (single class)	Post-hoc exploratory

Fault macro-F1 is calculated over the three fault-location classes within each severity subgroup; healthy recordings are reported separately and therefore have no fault macro-F1. * Post-hoc exploratory configuration identified after inspection of the same 3000-rpm holdout.

Because severity is inferred from recording identifiers and the subgroups are small, these results are descriptive only. The observed differences should not be interpreted as evidence that light-severity faults are intrinsically easier to diagnose than high-severity faults.

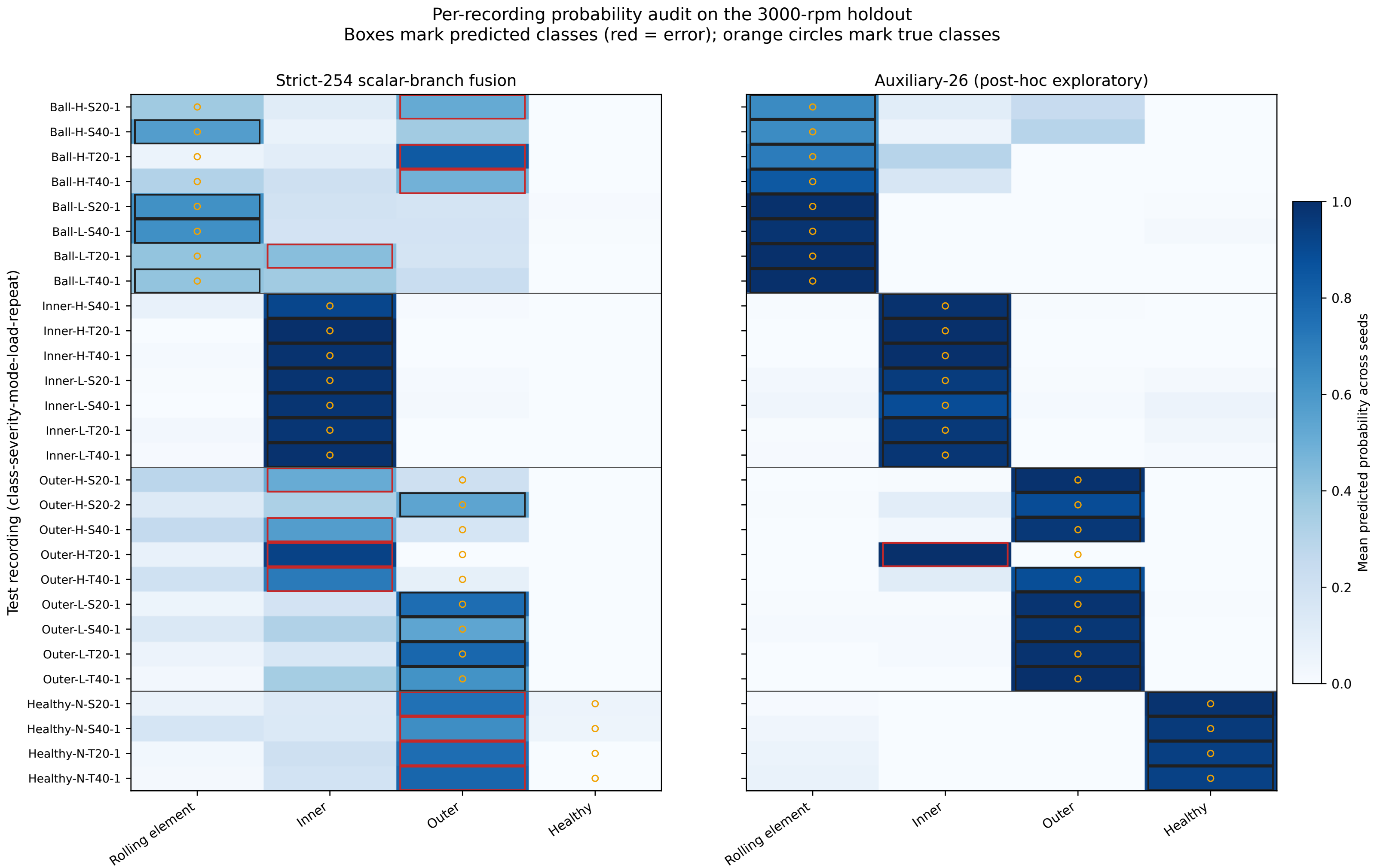

Figure S3: Per-recording class-probability audit on the 3000-rpm holdout, averaged across optimization seeds. Boxes mark the argmax class of the seed-averaged probability vector; red boxes denote errors, and orange circles denote the true class. Publication-facing class labels use "rolling element"; original recording identifiers retain the dataset prefix "Ball". The auxiliary-26 panel is post-hoc exploratory on the same holdout and is shown only to localize the associated recording-level behavior.

7 Recording-Level Bootstrap

Intervals resample complete test recordings within class for 2,000 replicates. They are conditional empirical resampling intervals for the listed finite test recordings and fitted seed. They neither quantify uncertainty for future physical specimens nor convert post-hoc model selection into confirmatory evidence.

Table S15: Class-stratified recording bootstrap for selected 3000-rpm models. Point estimates are recording-level macro-F1 values obtained after mean-probability aggregation within each recording. Conditional empirical intervals resample complete recordings within class and condition on the listed finite test set and trained seed; they are not intervals for future specimens. The Auxiliary-26 row is post-hoc exploratory, and the intervals do not remove its selection bias.

Model	Evidence status	Seed	Point macro-F1	95% interval	Recordings
Strict-254 scalar-branch fusion	Matched protocol	42	0.5348	[0.4242, 0.6244]	28
Strict-254 scalar-branch fusion	Matched protocol	43	0.2906	[0.2101, 0.3713]	28
Strict-254 scalar-branch fusion	Matched protocol	44	0.4478	[0.3439, 0.5519]	28
Strict-254 scalar-branch fusion	Matched protocol	45	0.2208	[0.1540, 0.2955]	28
Strict-254 scalar-branch fusion	Matched protocol	46	0.3359	[0.2189, 0.4581]	28
Auxiliary-26 branch*	Post-hoc exploratory	42	0.9374	[0.8453, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	43	0.9686	[0.9059, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	44	0.9686	[0.9059, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	45	0.9375	[0.8381, 1.0000]	28
Auxiliary-26 branch*	Post-hoc exploratory	46	0.9055	[0.7967, 1.0000]	28
Raw vibration-current CNN	Matched protocol	42	0.4367	[0.2922, 0.5515]	28
Raw vibration-current CNN	Matched protocol	43	0.8888	[0.7297, 1.0000]	28
Raw vibration-current CNN	Matched protocol	44	0.8444	[0.7134, 0.9375]	28

8 Model Architectures and Fixed Hyperparameters

Table S7 reports the publication-facing settings implemented in the formal benchmark. Classical hyperparameters were fixed before final test evaluation. Neural early stopping and checkpoint selection used only the frozen internal-validation recordings; held-out test recordings did not inform scaling, checkpoint selection, or hyperparameter choice.

Table S16: Model architectures, preprocessing, and fixed hyperparameters.

Model	Input and preprocessing	Architecture and fixed hyperparameters	Selection and size
Logistic regression	254 engineered fields; training-median imputation and StandardScaler	C=1.0; L2 penalty; lbfgs solver; max_iter=1000	Fixed before test; 1,020 coefficients/intercepts for 254 inputs and four classes
RBF-SVM	254 engineered fields; training-median imputation and StandardScaler	RBF kernel; C=3.0; gamma=scale	Fixed before test; support-vector count is training-data dependent
Random forest	254 engineered fields; training-median imputation; no scaling	200 trees; max_depth=None; min_samples_leaf=1; all CPU cores	Fixed before test; trained tree size is data dependent

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Table S16 continued

Model	Input and preprocessing	Architecture and fixed hyperparameters	Selection and size
MLP	254 engineered fields; training-median imputation and StandardScaler	254->128->4; ReLU; Adam; alpha=1e-4; max_iter=300	Fixed before test; 33,156 trainable weights and biases
XGBoost	254 engineered fields; training-median imputation; no scaling	200 trees; max_depth=4; learning_rate=0.05; subsample=0.9; colsample_bytree=0.9; multiclass log-loss	Fixed before test; trained tree size is data dependent
Raw CNN	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Conv1d 6->32 (kernel 9), batch normalization, ReLU, max-pool 4; Conv1d 32->64 (kernel 7), batch normalization, ReLU, global average pooling; linear 64->4	Best validation macro-F1; 16,612 trainable parameters
TCN	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Three non-residual dilated Conv1d blocks, width 32, kernel 5, dilations 1/2/4; batch normalization and ReLU; global average pooling; linear 32->4	Best validation macro-F1; 11,620 trainable parameters
Transformer	Six vibration-current channels; 8,192 samples; train-only per-channel z-score	Conv1d token projection 6->64 (kernel 9, stride 8); two encoder layers; d_model=64; four heads; feed-forward width 128; dropout=0.1; mean pooling; linear 64->4	Best validation macro-F1; 70,724 trainable parameters
Fusion family	Separate three-channel vibration and current sequences plus 0/2/26/28/254/262 scalar definitions; train-only sequence and scalar z-score	Each sequence encoder: Conv1d 3->32 (kernel 9), max-pool 4, Conv1d 32->64 (kernel 7), global average pooling. Scalar MLP d->32->32. Concatenated width 160, sigmoid gate 160->160, classifier 160->64->4, dropout=0.2	Best validation macro-F1; parameters: 68,420 / 68,452 / 69,220 / 69,284 / 76,516 / 76,772 in the listed scalar order

Classical configurations were fixed before final test evaluation. Neural checkpoints were selected by the highest internal-validation macro-F1. Neural models used class-weighted cross-entropy, Adam with learning rate 10^{-3}, batch size 64, at most 30 epochs, and patience 8; fusion models additionally used weight decay 10^{-4}.

9 Reproducibility Notes

The verified runtime used Python 3.12.13, PyTorch 2.11.0+cu128, and an NVIDIA GeForce RTX 5080. Automated tests cover train-only sequence and scalar standardization, constant channels, missing scalar values, serialization of fitted statistics, source-recording separation across all data roles, raw-file audit integrity, classical source-recording aggregation, and the evidence-status labels applied to exploratory configurations. The software supplement includes frozen split maps, a de-identified window-to-recording index, exact input schemas, compact raw/session audit tables, result tables, and reproduction commands; raw MCC5 signals are not redistributed.